ENERCO Energy Management System

Key Features & Capabilities Summary

Quick Reference
Solar • BESS • Factory Load

What is ENERCO?
A smart software platform that connects solar panels, large battery storage (BESS), factory electricity demand, and the power grid. It automatically stores cheap surplus solar power and uses battery power during expensive peak hours to lower electricity bills.

1. What It Analyzes (Real-Time Monitoring)

- 🌞 **Solar vs. Factory Power Balance:** Continuously tracks solar generation (MW) against factory power consumption to route surplus energy or cover shortages.
- 🔋 **Battery Storage (BESS) Health:** Monitors charge percentage (SoC), overall battery health (SoH), temperature, and keeps charge within safe limits (25% to 95%).
- 🔍 **39 Solar Panel Strings:** Monitors electricity output, temperature, and dust buildup across 3 plant sites (Chennai, Coimbatore, Madurai). Flags panels hotter than 45°C.
- 💰 **Electricity Tariffs & Savings:** Watches live electricity prices (off-peak ₹4.80 vs. peak ₹9.40/kWh) and calculates real-time money saved by avoiding expensive grid power.

2. What It Predicts (Smart Forecasting)

- 🌤️ **Tomorrow's Solar Power:** Ingests live weather forecasts (Open-Meteo) including sunlight, clouds, and temperature to predict next-day energy generation (MWh).
- ⚠️ **Equipment Overheating & Trips:** Warns operators up to 4 hours in advance if a solar inverter or battery cell climbs above 50°C, preventing hardware damage.
- ⚡ **Storms & Grid Blackouts:** Detects approaching bad weather or grid instability, automatically pre-charging the battery to keep the factory running without power cuts.

3 Sites	39 Panels	2–20 MWh	₹4.8–₹9.4
TAMIL NADU PLANTS	LIVE STRING SENSORS	BATTERY CAPACITY	TARIFF RATES / KWH

3. What It Visualizes (Screens & Graphs)

- 🔀 **Live Energy Flow Diagram:** Animated screen showing real-time power transfers between Solar Panels, Battery, Factory, and the Grid with live megawatt numbers.
- 📈 **5 Interactive Analytics Graphs:** Clear visual charts for (1) Solar vs. Load, (2) Battery Charge %, (3) Grid Import/Export, (4) Tariff Rates, and (5) Daily Money Saved.
- 🗺️ **Interactive Plant Map:** Map of 3 plant sites (Chennai, Coimbatore, Madurai) with color pins: Green (Normal), Amber (Warning), and Red (Fault).
- 📋 **30-Day Asset Health Curves:** Click any of the 39 panels to open an interactive 30-day health graph showing performance trends over time.

4. What It Reports (Downloadable Documents)

- 📄 **Daily Yield & Economics** `PDF` **:** Total solar power harvested, self-consumption %, peak power avoided, and daily rupee savings.
- 📄 **AI Optimizer Savings** `PDF` **:** Comparison of actual energy spend vs. what it would have cost without battery storage.
- 📊 **Battery Health Summary** `CSV` **:** Full battery audit including charge cycles, cell temperatures, and warranty records.
- 📄 **Monthly Performance (PR)** `PDF` **:** Standard solar plant efficiency score (IEC 61724 PR rating).
- 📊 **Maintenance & Alarm Log** `CSV` **:** Detailed list of all alarms, inverter trips, and operator fixes.

5. Other Key Features ("What-If" & AI)

- 🔮 **"What-If" Planning Simulator:** Use interactive sliders to adjust battery size (MWh) and solar size (MW). Instantly see total investment cost, payback period (ROI in years), and CO2 offset.
- 🤖 **Context-Aware AI Chatbot:** Ask questions like "*Which panel needs attention?*" or "*How much battery is available tonight?*" and get instant answers based on live sensor data.
- ⚡ **Smart Automation Modes:** Switch between **Autonomous** (auto-dispatch), **Ask Permission** (operator approval prompts), or **Manual** control.